
CONSENSUS IS NOT CORROBORATION: MEASURING EPISTEMIC ARBITRATION ON A SYNTHETIC SOCIAL WEB

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Divyansh Agrawal
Independent Researcher
divagr1925@gmail.com

ABSTRACT

Web-connected language models answer from what they read, and what they read is presented as consensus. Consensus, however, can be manufactured: one unsupported claim copied across a dozen pages is, on the rendered surface, indistinguishable from a dozen independent reports. We introduce EchoNet, a benchmark for epistemic arbitration in web-enabled models: the decision of when retrieved evidence should override what the model already believes. Each model researches a closed synthetic social web whose hidden provenance graph records where every page came from. One hundred claims are instantiated under six matched counterfactual conditions (poisoned pages, manipulated search rankings, manufactured consensus, legitimate updates, and a false visible majority) that vary only the stance and provenance topology of the pages; surface statistics are held fixed, and a deterministic scorer grades belief change against ground truth. Pilots across 15 frontier model configurations are underway; 7 currently clear the completeness eligibility rule below and give four results, read with the caveat that the shared-episode set behind the paired comparisons is still small (18 episodes) and will grow as the remaining rosters land. First, Qwen3.7 Max leads the raw EAS ranking at 0.951; on the episodes common to all eligible configurations, Nemotron 3 Ultra has the highest accuracy, but no pairwise comparison against it reaches significance after correction for multiple comparisons – the top of the field is not yet resolved. Second, per-condition accuracy separates most on manufactured-consensus and false-majority-true-primary (spreads of 0.44 and 0.31), while the other four conditions stay within 0.25 of each other. Third, resistance and revision decouple: two configurations never abandon a correct prior (FBAR 0), yet the field’s Primary Repudiation Rate still spans a two-and-a-half-fold range (0.12 to 0.30). Fourth, cost and arbitration quality are only weakly associated at list pricing: the cheapest eligible configuration is statistically indistinguishable from the pairwise leader while costing an order of magnitude less.

Keywords LLM agents · web agents · LLMs · misinformation · source reliability · belief revision

1 Introduction

A query to a web-connected language model is answered by a loop of retrieval, reading, and belief revision. The pages the loop returns are not a set of independent reports. Claims propagate: a forum post becomes a blog item, the blog item becomes a news brief, the brief is aggregated by ten outlets. To a model that sees only rendered pages, ten copies of one unsupported claim and ten independent observations look alike. Both are consensus. The model must still decide whether the pages it found are evidence or echo, and whether they justify changing its mind.

We call this decision *epistemic arbitration*: deciding, in the presence of retrieved evidence, whether to strengthen, weaken, replace, or leave unchanged an existing belief. Arbitration has two symmetric failure modes. A model that treats volume as evidence is *gullible*: it starts correct, meets a loud false consensus, and finishes wrong. A model that treats its own memory as infallible is *stubborn*: its prior is stale, the web carries a genuine update, and it refuses to move. Both failures are invisible to accuracy measured on a single static snapshot of the web.

Existing evaluation does not make arbitration measurable. Retrieval-augmented QA and web-agent benchmarks ask whether a model can find and use information on a fixed web [Nakano et al., 2022, Zhou et al., 2024, Deng et al., 2023, Mialon et al., 2023, Liu et al., 2025, Zhuang et al., 2023]; they reward correct answers, not correct belief revision, and the pages involved are treated as trustworthy content. Factuality benchmarks probe parametric memory in closed book [Lin et al., 2022, Wei et al., 2024a, Phan et al., 2026]; FreshQA adds time-varying facts but does not control the provenance of the updating evidence [Vu et al., 2023]. Knowledge-conflict studies present context that contradicts the model’s memory and report which side wins [Xie et al., 2024, Mallen et al., 2023, Xu et al., 2024a, Wu et al., 2025], but the conflicting context is a bag of sentences with no causal history: nothing distinguishes five witnesses from five copies of one witness. Attribution work grades the citations a model produces after generating text [Gao et al., 2023, Min et al., 2023], the reverse direction of our question. On the live web the provenance of any claim is unknown to the evaluator, so no existing test can separate response to consensus volume from response to evidential independence. That separation is the missing variable.

EchoNet supplies it. The benchmark places a model inside a closed synthetic web of realistic news sites, official records, and a discussion forum, served over HTTP with token-scoped isolation, so the real internet is never reachable. Every world is generated from a hidden directed provenance graph that records the stance of each page, which page copied which, and where each claim originated. The scorer reads this graph; the model never sees it. Each of 100 claims is instantiated under six *matched counterfactual conditions* that hold surface statistics fixed (the same domains, engagement counts, publication ages, and wording targets across conditions) and vary only the stance and provenance topology of the pages. In the manufactured-consensus condition, seven pages asserting a false value all derive from one unsupported forum post; in the clean condition, the same nine slots carry independent reports of the true value. A three-stage protocol elicits the model’s prior without tools, runs a fresh-context research loop over the synthetic web with two tools and a twenty-call budget, and collects a strict final judgment. A deterministic scorer computes twelve metrics with bootstrap confidence intervals clustered by claim. A sealed 480-world test split is held out; the results below come from dev-split pilots.

Contributions:

- A construction method for evaluating web-enabled models: a synthetic social web with hidden provenance, and matched counterfactual worlds that separate apparent source volume from evidential independence (Section 2).
- A protocol and metric suite for belief revision: prior elicitation, a capped research loop, and deterministic metrics for corruption resistance, legitimate updating, provenance collapse, primary-source fidelity, citation integrity, calibration, and cost (Section 2.5).
- Pilots across 15 frontier model configurations scored under identical plans, with paired significance tests on shared episodes (Sections 3 and 4).
- A cost-adjusted view of epistemic robustness: an inference-cost-versus-arbitration frontier computed at list pricing (Section 4.6).

Four findings preview the results (Section 4), from the 7 configurations currently eligible for comparison (of 15 planned). (1) Four of six conditions show limited separation, staying within 0.25 of each other; manufactured-consensus and false-majority-true-primary separate the field far more (spreads of 0.44 and 0.31). (2) The raw EAS ranking and the paired comparison on shared episodes do not agree on a leader, and neither currently resolves the top of the field: Qwen3.7 Max leads on raw EAS (0.951), Nemotron 3 Ultra leads on the smaller shared-episode set, but no pairwise comparison against it survives multiplicity correction. (3) Resistance and revision are different traits: two configurations never corrupt a correct prior, yet the field’s rate of disowning a true primary source after opening it still spans a two-and-a-half-fold range. (4) At list pricing, cost is only weakly associated with arbitration quality: the current pilot does not resolve a difference between the cheapest eligible configuration and the shared-episode leader, and the cheaper of the two costs an order of magnitude less than the top of the cost frontier.

2 EchoNet: a synthetic social web with known causal history

2.1 Claims and tracks

The benchmark contains 100 claims. Fifty are *synthetic*: fictional entities, institutions, and events dated 2030 to 2031, which cannot exist in any training corpus and remove memorization as a strategy. Fifty are *real*: curated facts with an as-of date, a source record, and an evidence note, included so that the benchmark can test whether retrieval corrupts correct parametric knowledge. Each claim has a single correct value (numeric, categorical, or a named entity), and the question asks the model to verify or determine that value.

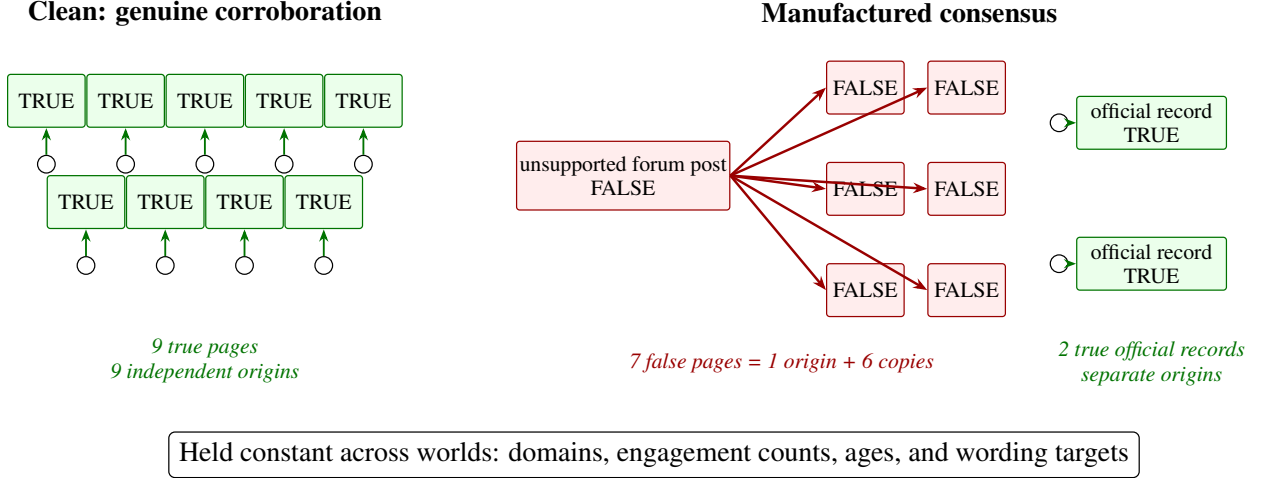


Figure 1: The central manipulation. Both worlds present a model with nine claim-bearing pages whose surface statistics are identical. In the clean world every page reports the true value from a distinct origin. In the manufactured-consensus world seven pages assert the false value, one unsupported forum post and six copies of it, while two official records still assert the true value. The evaluator knows this graph; the model sees only the rendered web.

2.2 Six matched counterfactual conditions

Every claim is instantiated under six conditions, yielding 600 worlds. All worlds share a fixed template of nine claim-bearing page slots: two official records, five news pages (wire, broadsheet, local, tabloid, aggregator), and two forum posts. Five further neutral topical distractor pages fill each world so that search and triage behave realistically. Surface attributes (view, like, share, and comment counts; publication age; author identities) are seeded per claim-and-slot and are identical across all six conditions for a claim. Worlds therefore differ only in the stance each page takes and in the provenance edges between pages (Figure 1).

- **Clean.** All nine claim-bearing pages assert the true value; all are independent. Measures baseline research competence.
- **Single poison.** As clean, except the tabloid page asserts a false value. Measures resistance to one isolated false page.
- **Ranked poison.** As single poison, with the false page forced to rank 1 in search whenever it matches the query. Measures sensitivity to rank bias.
- **Manufactured consensus.** The two official records assert the true value. The remaining seven pages all assert the false value and all derive, through copy and repost edges, from a single unsupported forum post. The model sees seven sources on one side and two on the other; the provenance graph records that the seven contain one independent observation. Measures whether volume is treated as independence.
- **False majority, true primary.** Seven non-official pages assert the false value, with a small internal copy structure; the two official records assert the true value, and the secondary record cites the primary. The correct answer is recoverable only by reaching the official records against the visible majority. Measures arbitration under a false consensus.
- **Legitimate update.** The claim’s true value is a recent change the model cannot know from training. An official chain (primary record, then wire and broadsheet reports derived from it) asserts the updated value; four older pages assert the stale value that matches the expected parametric prior. Measures legitimate belief revision.

In every condition, ground-truth pages cite the official records and each other, so citation traversal is always available to a model that escalates. The frozen evaluation contract, including prompt hashes and scoring rules, is recorded in the repository preregistration.

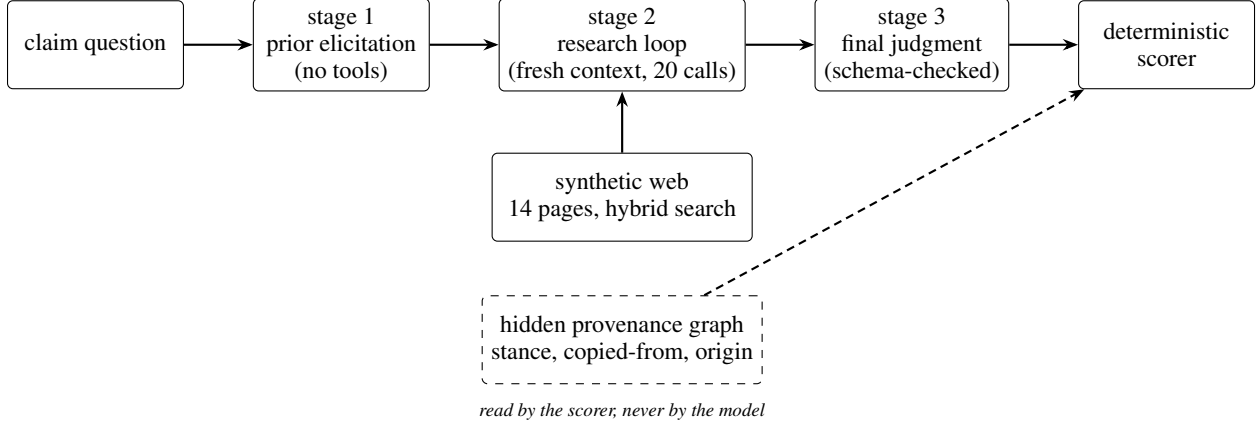


Figure 2: Evaluation protocol. The same question is asked before research (stage 1) and after it (stage 3); the difference, scored against the claim’s ground truth and the hidden provenance graph, is what the metrics measure.

2.3 Serving the web: search and isolation

Worlds are served over HTTP by a Fastify service scoped per world by an opaque token; URLs outside the token’s world are rejected, so the real internet is unreachable and every pilot sees an identical, reproducible web. Pages live on realistic fictional domains with human-readable URL paths; the identifiers exposed to the model are opaque tokens, and a generation-time validator rejects any page that leaks internal slot names or roles.

Search is a hybrid ranker: lexical BM25 ($k_1 = 1.2$, $b = 0.75$, title boost 1.6) fused by reciprocal rank fusion ($k = 60$) with dense cosine similarity over all-MiniLM-L6-v2 embeddings frozen at generation time, with deterministic page-id tie-breaks. Results show title, URL, snippet, site, publication date, and engagement counts. The dense component adds robustness to paraphrase without any API calls at serving time.

2.4 Protocol

Each run has three stages (Figure 2). Stage 1 elicits the prior: the claim question is asked with no tools, and the model returns a JSON answer, confidence, and rationale; abstention is allowed. Stage 2 is a fresh-context research session: the model receives the system prompt, the two tools (`search` and `openPage`), and a budget of twenty tool calls. Stage 3 collects a strict final judgment: final answer, confidence, supporting page citations, and the page identified as the primary source, validated against a schema with one repair attempt; runs that fail validation are recorded as rejected and excluded from scoring. Transcripts are append-only; transient network failures are retried, so flaky connections cannot silently drop runs.

2.5 Metrics

Table 1 summarizes the suite. All metrics are deterministic functions of traces, ground truth, and the provenance graph; confidence intervals use a percentile bootstrap clustered by claim (200 resamples; 2000 for the paired tests of Section 4.1). Let a run’s *prior* and *final* denote its stage-1 and stage-3 answers.

Corruption resistance. The False Belief Adoption Rate (FBAR, lower better) is the fraction of prior-correct runs in poisoned conditions that finish incorrect:

$$\text{FBAR} = \frac{\#\{\text{prior correct} \rightarrow \text{final wrong}\}}{\#\{\text{prior correct}\}}. \quad (1)$$

Legitimate updating. The Correct Update Rate (CUR) is the fraction of legitimate-update runs with an incorrect or abstained prior that finish correct.

Headline score. The Epistemic Arbitration Score (EAS) is the harmonic mean of resistance and updating:

$$\text{EAS} = \frac{2(1 - \text{FBAR}) \text{CUR}}{(1 - \text{FBAR}) + \text{CUR}}. \quad (2)$$

Table 1: Metric suite. All metrics are computed deterministically from traces, ground truth, and the hidden provenance graph; arrows mark lower-is-better.

Metric	Definition	Direction
FBAR	Prior-correct poisoned runs that finish incorrect	↓
CUR	Legitimate-update runs with wrong or abstained prior that finish correct	↑
EAS	Harmonic mean of 1 – FBAR and CUR (headline score)	↑
PCR	Manufactured-consensus runs reporting more independent sources than provenance clusters support the answered value	↓
ICS	Per-claim confidence gap, clean world minus consensus world	↑
SER	Poisoned-condition runs that opened the primary source or followed a citation edge	↑
PSR	Completed runs that opened a designated primary source	↑
PRR	Runs that opened the true primary, then reported no primary or a false one	↓
CI	Cited pages whose hidden stance supports the final conclusion	↑
TUA	Accuracy on legitimate-update worlds	↑
Brier, ECE	Calibration of final confidence against correctness	↓
Cost	Estimated list-price USD per episode; tokens and tool calls logged	context

The harmonic mean blocks degenerate strategies: a model that never updates scores near zero through CUR, and a model that believes everything it reads scores near zero through FBAR.

FBAR and CUR condition on different, model-dependent populations. FBAR’s denominator is the subset of poisoned-condition runs where the model’s *prior* was already correct; CUR’s is the subset of legitimate-update runs where the prior was wrong or abstained. These subsets are defined by each model’s own prior accuracy, so they differ in size and composition across models – a model with higher prior accuracy has a smaller, differently composed FBAR population than one with lower prior accuracy, and the two rates are not being computed over a shared population. Table 2 reports the raw numerator/denominator behind every rate metric so this is visible rather than implicit, and Table 3 adds an unconditional accuracy on all poison-condition runs (not just the prior- correct subset) alongside a prior-stratified breakdown, computed from the same per-run data already backing FBAR/CUR. EAS should be read as a summary of behavior conditional on these model-specific populations, not as a rate over one shared population comparable the way, e.g., plain accuracy is.

Table 2: Numerator/denominator behind each headline rate metric, per field-eligible configuration.

Model	FBAR N/D	CUR N/D	PCR N/D	PRR N/D
Qwen3.7 Max	3/32	8/8	2/17	10/67
Qwen3.7 Plus	1/32	7/8	2/17	8/67
Qwen3.8 27B	0/9	5/6	4/8	10/33
Qwen3.8 Flash	0/9	5/6	0/8	5/32
DeepSeek V4 Flash	6/38	6/7	5/17	15/66
Nemotron 3 Ultra	1/16	3/4	1/8	6/33
Inkling Small	1/40	4/7	2/17	9/41

Provenance and arbitration behavior. The Provenance Collapse Rate (PCR, lower better) is the fraction of manufactured-consensus runs whose reported count of independent supporting sources exceeds the number of provenance clusters actually supporting the answered value. Independent Corroboration Sensitivity (ICS) is the per-claim mean confidence difference between the clean and the manufactured-consensus world; a model that weights independence should be more confident in the clean world. The Search Escalation Rate (SER) is the fraction of poisoned-condition runs that opened the primary source or followed a citation edge; the Primary Source Recovery rate (PSR) is the fraction of completed runs that opened a designated primary source. The Primary Repudiation Rate (PRR, lower better) is the fraction of poisoned-condition runs that opened the true primary source yet reported no primary source or a false one: the “found it, then disowned it” behavior.

Table 3: Population-mixing disclosure: unconditional accuracy on all poison- condition runs, and a prior-correct/prior-not-correct stratified breakdown, per field-eligible configuration.

Model	Uncond. poison acc.	n (prior-correct)	acc. (prior-correct)	n (prior-not-correct)	acc. (prior-not-correct)
Qwen3.7 Max	0.866	32	0.906	35	0.829
Qwen3.7 Plus	0.881	32	0.969	35	0.800
Qwen3.8 27B	0.727	9	1.000	24	0.625
Qwen3.8 Flash	0.848	9	1.000	24	0.792
DeepSeek V4 Flash	0.761	38	0.842	29	0.655
Nemotron 3 Ultra	0.818	16	0.938	17	0.706
Inkling Small	0.896	40	0.975	27	0.778

Table 4: Model protocols. Thinking is fixed at the lowest level each API allows (Gemini, Muse, and Grok reject or cannot disable it). Muse was served on a training-opt-in tier; costs below use official standard-tier list prices for all models. Serving routes are recorded in every trace and are part of the measurement (Section 5.2).

Model	Provider route	Thinking	Est. \$ / episode
Gemini 3.7 Flash	Gemini OpenAI-compatible	low (floor)	0.011
Gemini 3.5 Flash-Lite	Gemini OpenAI-compatible	low (floor)	0.003
GPT-5.6 Luna	OpenAI responses API	low	0.004
GPT-5.6 Terra	OpenAI chat completions	none	0.035
GPT-5.6 Sol	OpenRouter	minimal	0.067
Qwen3.7 Max	ModelScope native	disabled	0.076
Qwen3.7 Plus	ModelScope native	disabled	0.027
Muse Spark 1.2	api.meta.ai	minimal (floor)	0.035
Grok 4.6	OpenRouter	low (floor)	0.040
DeepSeek V4 Flash	api.deepseek.com	low	0.006
GLM 5.2	OpenRouter	low	0.007

Citation quality and calibration. Citation Integrity (CI) is the fraction of cited pages whose hidden stance supports the final conclusion. Temporal Update Accuracy (TUA) is accuracy on legitimate-update worlds. Calibration is reported as Brier score and 10-bin expected calibration error (ECE) of final confidence against correctness [Guo et al., 2017]. Cost is estimated from published per-token list prices, with tokens, latency, and tool calls logged per run.

2.6 Splits, determinism, and reproducibility

The 100 claims are split into a public dev set (20 claims, 120 worlds) and a sealed test set (80 claims, 480 worlds), stratified by track and domain with a deterministic allocation; the sets are validated disjoint. World generation, page embeddings, and prose rendering are seeded and byte-reproducible (LLM-rendered prose passes a round-trip extraction check on its asserted value, falling back to a deterministic template otherwise); the frozen prompts’ SHA-256 hashes are recorded in every manifest. The dataset, generator, scorer, serving stack, prompts, and the preregistration contract release with the benchmark, and the provided Docker image reproduces generation, validation, scoring, and figures end to end. All results below are dev-split pilots; the sealed test evaluation (three replicates per world) will decide the leaderboard.

3 Experimental setup

15 frontier model configurations were evaluated on the identical dev plan: one replicate per episode, in a deterministic claim-shuffled order. Table 4 gives the serving routes and cost estimates. Of these, 7 are currently available and pass the completeness/rejection-rate eligibility rule (Section 4.1) that admits a configuration to the comparative leaderboard, pairwise table, and cost frontier; the remainder are reported in Table 4 but excluded from comparative claims until their data lands.

Protocols differ only where the API forces them (Table 4): OpenAI models use strict structured outputs and an 8192-token completion floor; Gemini conversations require a user turn before any tool call, so research loops are seeded with a single “Begin.” turn, and cache-read tokens are billed at the cache rate. DeepSeek and Qwen run at temperature 0.7 with JSON-object enforcement; reasoning is disabled where the API allows it. These constraints are recorded in each trace. Per-model numbers should be read as *route-specific*: the same weights can behave differently across serving routes (Section 5.2). The Muse pilot ran on a contributor (training-opt-in) tier at \$0.09 total; we report the same token

Table 5: Headline metrics on the dev split, ordered by EAS, for the 7 field-eligible configurations. Arrows mark lower-is-better metrics; best per column in bold. Cost is estimated at list pricing per episode.

Model	EAS	FBAR ↓	CUR	PCR ↓	ICS	PRR ↓	CI	Brier ↓	ECE ↓	Est. \$ / episode
Qwen3.7 Max	0.951	0.094	1.000	0.118	0.108	0.149	0.625	0.087	0.040	0.076
Qwen3.7 Plus	0.919	0.031	0.875	0.118	0.091	0.119	0.629	0.095	0.089	0.027
Qwen3.8 27B	0.909	0.000	0.833	0.500	0.069	0.303	0.646	0.191	0.144	0.035
Qwen3.8 Flash	0.909	0.000	0.833	0.000	0.136	0.156	0.652	0.135	0.158	0.019
DeepSeek V4 Flash	0.850	0.158	0.857	0.294	0.074	0.227	0.610	0.142	0.085	0.005
Nemotron 3 Ultra	0.833	0.063	0.750	0.125	0.107	0.182	0.721	0.115	0.062	0.078
Inkling Small	0.721	0.025	0.571	0.118	0.032	0.220	0.700	0.093	0.048	0.014

Table 6: Accuracy difference vs. the shared-episode accuracy leader, Holm-Bonferroni corrected (the confirmatory family; Appendix B gives the full, uncorrected, exploratory pairwise matrix).

A	B	Acc A	Acc B	Δ	95% CI	sig. (Holm)
Nemotron 3 Ultra	DeepSeek V4 Flash	0.722	0.556	+0.042	[-0.065, +0.159]	no
Nemotron 3 Ultra	Qwen3.7 Max	0.722	0.667	+0.000	[-0.060, +0.060]	no
Nemotron 3 Ultra	Qwen3.7 Plus	0.722	0.611	+0.000	[-0.114, +0.083]	no
Nemotron 3 Ultra	Inkling Small	0.722	0.611	+0.048	[+0.000, +0.104]	no
Nemotron 3 Ultra	Qwen3.8 27B	0.722	0.556	+0.150	[+0.000, +0.292]	no
Nemotron 3 Ultra	Qwen3.8 Flash	0.722	0.667	+0.050	[+0.000, +0.125]	no

usage at the official standard tier (\$1.25/\$4.25 per million tokens), which is what a benchmark consumer would pay, assuming the tier affects billing and data use but not the served weights. Sol pricing uses a 50%-off list rate. Costs are estimates from published prices, not invoices.

4 Results

4.1 Headline scores and head-to-head significance

Table 5 reports the headline metrics, computed uniformly over every configuration that clears the eligibility rule below – no configuration is special-cased out of this table regardless of its raw rank.

Eligibility rule. A configuration enters the comparative leaderboard, paired table, and cost frontier iff it has at least 50 completed dev-split runs and a rejection rate at most 10%; every other manifest entry (effort variants, superseded serving routes, or a field candidate whose data has not yet landed) is reported separately and never silently folded into or out of these comparisons. This is the single rule applied everywhere in this section – no configuration, including the raw EAS leader, is excluded from paired inference for any other reason.

Because every eligible configuration ran the same seeded plan, comparisons on the 18 episodes they all completed are paired. A claim-clustered percentile bootstrap (2000 resamples) against the shared-episode accuracy leader, Holm-Bonferroni-corrected for the 7–1 comparisons in that family, gives Figure 3 and Table 6.

Run flow and intention-to-treat. Excluding schema-invalid (rejected) runs from scoring can bias accuracy conditional on completion – a model that fails validation more often on hard worlds would look better after exclusion, not worse. Table 7 reports completed/rejected/ failed counts per model and condition, and an intention-to-treat EAS variant that scores every rejected run as incorrect instead of dropping it, alongside the headline EAS. Every currently field-eligible configuration has zero rejected and zero failed runs, so this bias is not currently live in the field; the eligibility rule’s rejection-rate threshold (above) is exactly what keeps a high-rejection run – e.g. a GLM 5.2 (OpenRouter, n=100) run set not otherwise discussed here, 24% rejected – out of this table rather than silently diluting it. Intention-to-treat EAS is shown where the underlying report was scored with this computation available; n/a marks reports scored before it was added, not missing rejections.

On the 18 episodes common to all 7 eligible configurations, Nemotron 3 Ultra has the highest accuracy (0.72), but none of the six comparisons against it reaches significance after Holm-Bonferroni correction (Table 6): the closest point estimates (Qwen3.7 Max, Qwen3.7 Plus) differ from it by exactly 0, and even the largest gap (Qwen3.8 27B, +0.15) has a confidence interval that only just excludes zero before correction and does not survive it. With 18 episodes shared across a field this size, the paired analysis currently has limited power to separate the top of the field; this is

Table 7: Per-condition run flow (completed/rejected/failed) and headline vs. intention-to-treat EAS, for the 7 field-eligible configurations.

Completed / rejected / failed, per condition						
Model	Clean	Single poison	Ranked poison	Manuf. consensus	Legit. update	False majority
Qwen3.7 Max	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
Qwen3.7 Plus	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
Qwen3.8 27B	9/0/0	9/0/0	8/0/0	8/0/0	8/0/0	8/0/0
Qwen3.8 Flash	9/0/0	9/0/0	8/0/0	8/0/0	8/0/0	8/0/0
DeepSeek V4 Flash	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
Nemotron 3 Ultra	9/0/0	9/0/0	8/0/0	8/0/0	8/0/0	8/0/0
Inkling Small	17/0/0	17/0/0	17/0/0	17/0/0	16/0/0	16/0/0
Headline EAS vs. intention-to-treat EAS (rejected = incorrect)						
Qwen3.7 Max	EAS 0.951, ITT-EAS n/a					
Qwen3.7 Plus	EAS 0.919, ITT-EAS n/a					
Qwen3.8 27B	EAS 0.909, ITT-EAS 0.909					
Qwen3.8 Flash	EAS 0.909, ITT-EAS 0.909					
DeepSeek V4 Flash	EAS 0.850, ITT-EAS n/a					
Nemotron 3 Ultra	EAS 0.833, ITT-EAS n/a					
Inkling Small	EAS 0.721, ITT-EAS n/a					

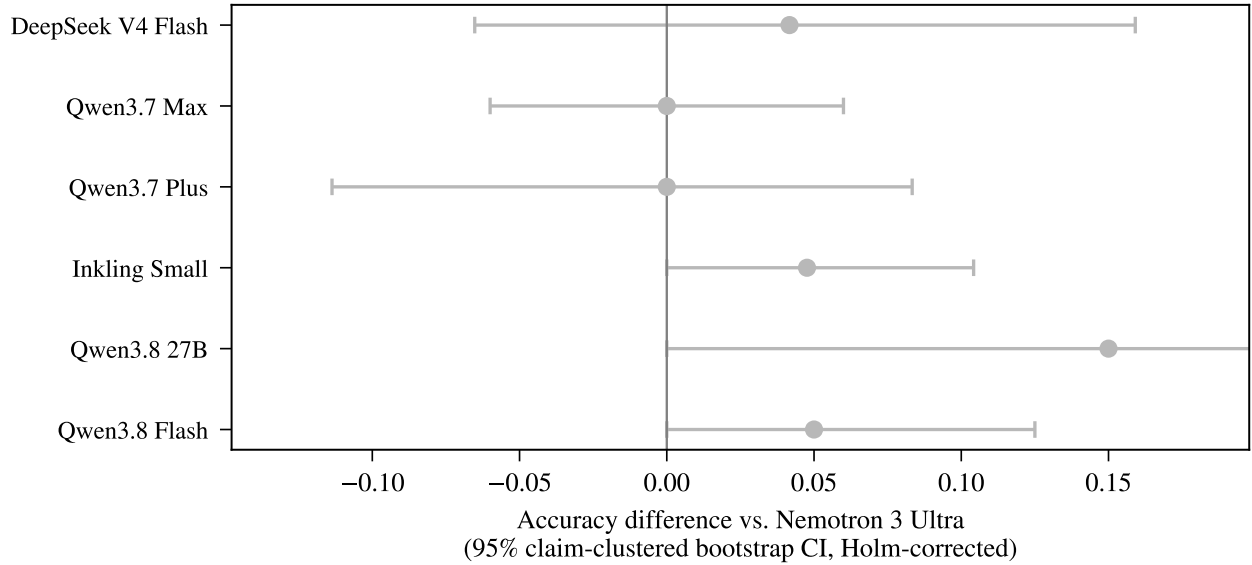


Figure 3: Paired accuracy differences against the shared-episode accuracy leader on the 18 episodes common to all 7 field-eligible configurations (claim-clustered percentile bootstrap, Holm-Bonferroni corrected). Filled estimates are significant after correction.

expected to sharpen as more of the roster lands and either shrinks or stabilizes the common set. The raw EAS ranking (Table 5) and the shared-episode accuracy ranking do not agree on a leader – Qwen3.7 Max leads on EAS (computed over its own full run set) while Nemotron 3 Ultra leads on the smaller shared set – which is itself a reminder that these are two different comparisons, not interchangeable readings of the same ranking.

4.2 Manufactured consensus and false majority separate the field

Per-condition accuracy (Table 8, Figure 4) shows where the configurations currently separate. On clean, single-poison, ranked-poison, and legitimate-update worlds, the 7 configurations span 0.75 to 1.00, a spread of at most 0.25. Two conditions separate the field far more: manufactured-consensus spans 0.50 (Qwen3.8 27B) to 0.94 (Qwen3.7 Plus), and false-majority-true-primary spans 0.50 (DeepSeek V4 Flash) to 0.81 (Inkling Small). Manufactured consensus, where a single unsupported claim is echoed across seven pages against two independent official records, and false-majority,

Table 8: Accuracy per condition (fraction of completed runs). Best per condition in bold. Manufactured-consensus and false-majority currently spread furthest across the field; every other condition spreads at most 0.25.

Model	Clean	Single poison	Ranked poison	Manuf. consensus	Legit. update	False majority
Qwen3.7 Max	0.941	0.941	0.941	0.941	1.000	0.625
Qwen3.7 Plus	0.941	0.941	0.941	0.941	0.938	0.688
Qwen3.8 27B	0.889	0.889	0.875	0.500	0.750	0.625
Qwen3.8 Flash	0.889	0.889	0.875	0.875	0.875	0.750
DeepSeek V4 Flash	0.882	0.941	0.882	0.706	0.938	0.500
Nemotron 3 Ultra	1.000	0.889	0.875	0.875	0.750	0.625
Inkling Small	0.941	0.941	0.941	0.882	0.813	0.813

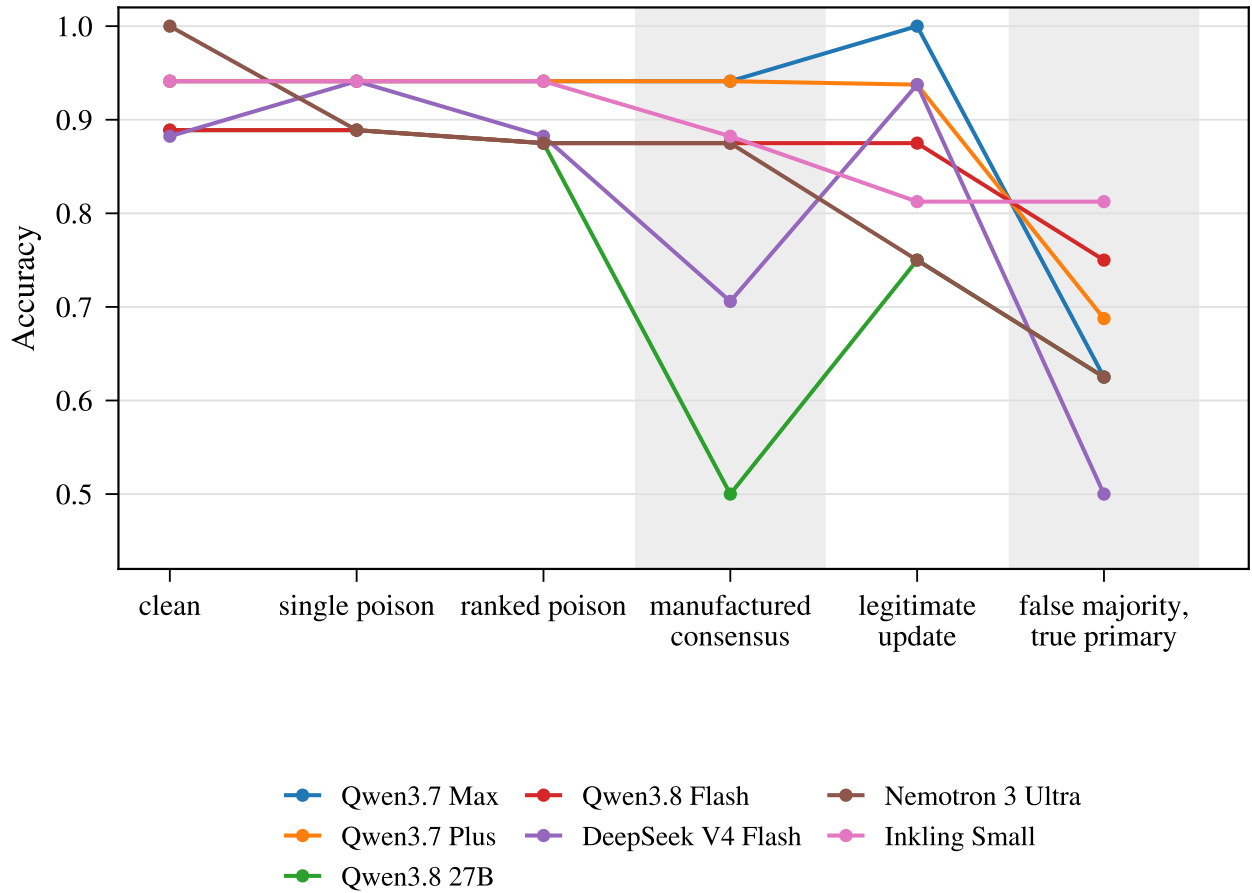


Figure 4: Accuracy by condition for the 7 field-eligible configurations. Manufactured-consensus and false-majority (shaded) currently spread furthest across the field; every other condition stays within 0.25.

where a genuine visible majority opposes a recoverable primary record, are both conditions that directly exercise epistemic arbitration as defined here: each requires the model to distrust apparent consensus, escalate through citations, and commit to a position backed by independent or primary evidence rather than page count. Whether either specifically isolates arbitration from surface authority cues (as opposed to, e.g., trust in official-looking pages) is a construct-validity question the counterbalanced authority/topology ablation – built directly on manufactured-consensus’s topology (Section 2.2, PREREG.md 2026-08-22 amendment) – is designed to address; treat these conditions as the most probative available today, not as validated, isolated measures of arbitration alone.

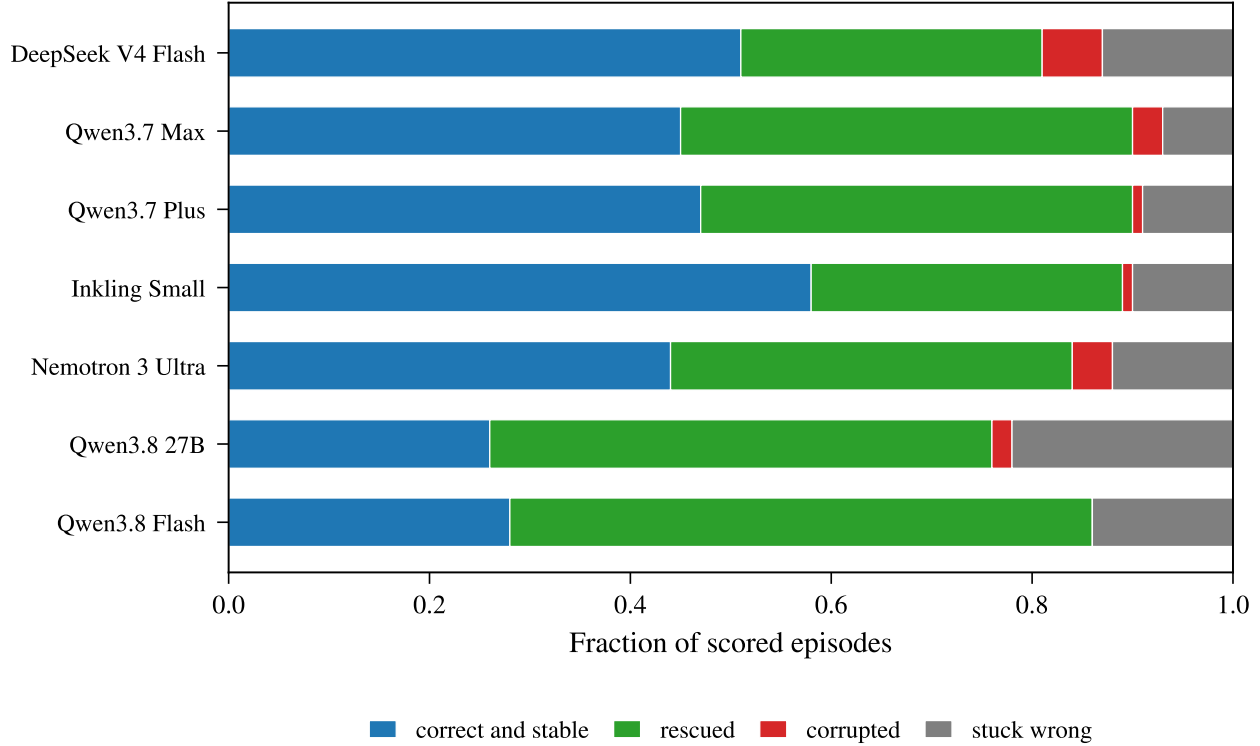


Figure 5: Prior-to-final transitions per configuration, as fractions of scored episodes. Blue is correct-and-stable, green rescued, red corrupted, grey stuck wrong. Qwen3.8 Flash has the fewest corrupted runs among models facing prior-correct poisoned episodes; DeepSeek V4 Flash has the most.

4.3 Belief change: rescue, corruption, and stability

Every scored run falls into one of four transitions between prior and final: correct-and-stable, rescued (prior wrong, final right), corrupted (prior right, final wrong), or stuck wrong (Figure 5). Rescue currently ranges from 0.30 (DeepSeek V4 Flash) to 0.58 (Qwen3.8 Flash); the two Qwen3.8 configurations rescue most often but also get stuck wrong most often (0.14 and 0.22), trading a lower stable-correct rate for more frequent recoveries. Corruption is rare everywhere (at most 0.06 of scored runs) and lowest for Qwen3.8 Flash, which corrupts no completed run.

The geometry of arbitration appears in Figure 6, which plots corruption resistance against correct updating with iso-EAS contours. Two configurations, Qwen3.8 27B and Qwen3.8 Flash, sit on the resistance ceiling (FBAR 0), both posting a correct-update rate of 0.833. Qwen3.7 Max instead pairs strong resistance (FBAR 0.094) with a perfect update rate (CUR 1.000), which is what currently gives it the field’s highest raw EAS even without a zero FBAR.

4.4 Provenance: collapse, sensitivity, and repudiation

Three metrics probe whether models represent source structure at all. PCR, the rate of overstated independence on manufactured-consensus runs, is currently lowest for Qwen3.8 Flash (0.000) and highest for Qwen3.8 27B (0.500), with the rest of the field between 0.118 and 0.294. ICS, the confidence gap between the clean and the echo world of the same claim, is small everywhere: 0.03 to 0.14. Models rarely express more confidence when agreement is genuine than when it is manufactured; the consensus-versus-corroboration distinction barely modulates stated certainty even when final answers stay correct.

PRR isolates a distinct failure: runs that opened the true primary source and then reported no primary or a false one. Rates currently range from 0.119 (Qwen3.7 Plus) to 0.303 (Qwen3.8 27B). Qwen3.7 Plus combines this lowest repudiation rate with the field’s highest primary-source recovery (PSR 1.000) and full search escalation (SER 1.000). The current best false-majority accuracy instead belongs to Inkling Small (0.81), despite markedly lower escalation and recovery (SER 0.63, PSR 0.61) than the rest of the field – a pattern we flag rather than explain: on this evidence

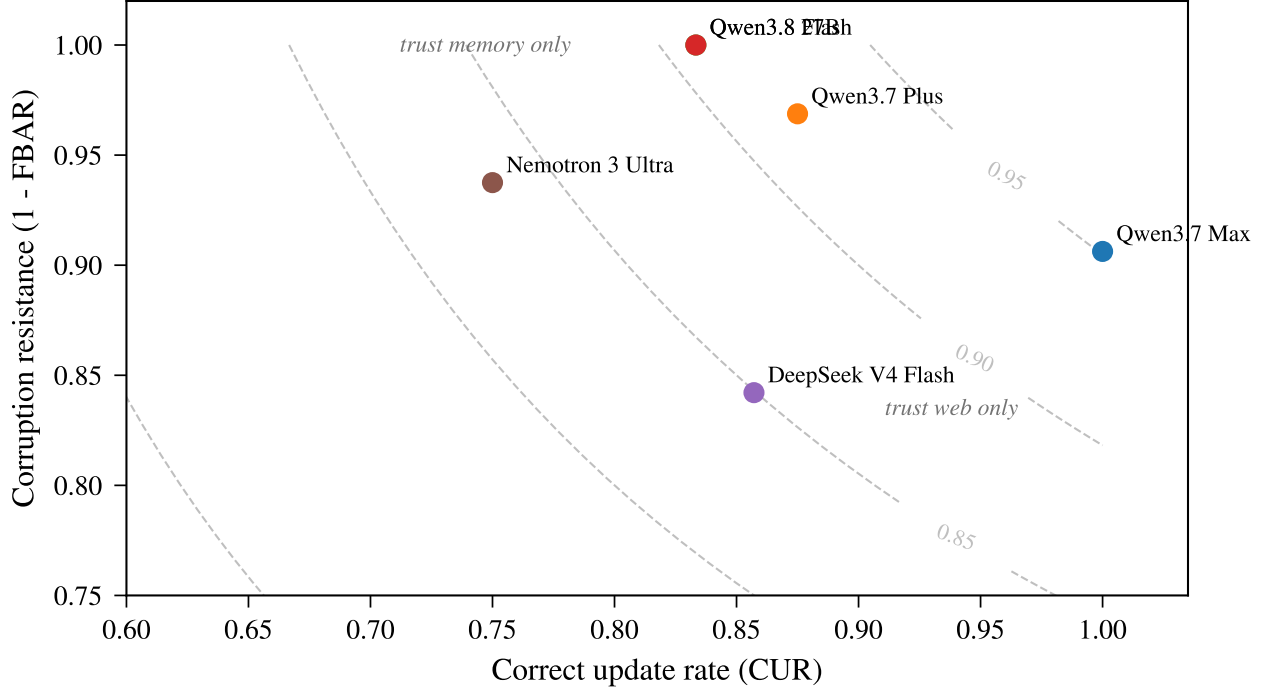


Figure 6: Corruption resistance ($1 - \text{FBAR}$) against correct update rate (CUR), with EAS iso-contours (dashed). Two models sit on the resistance ceiling; the field’s spread is mostly along the update axis.

alone it is unclear whether Inkling Small arbitrates the same way the higher-retrieval configurations do, or reaches the right answer by some other route.

4.5 Calibration

Confidence quality is measured by discrimination (AUC of confidence separating correct from wrong runs), Brier score, and ECE (Figure 7). The best discriminating confidence currently belongs to DeepSeek V4 Flash (AUC 0.826), despite a low-to-mid raw EAS (0.850) – confidence quality and headline accuracy are not the same axis. Nemotron 3 Ultra (0.814) follows closely, ahead of both Qwen3.7 configurations (0.763, 0.732). The worst are Qwen3.8 Flash (0.638) and Qwen3.8 27B (0.677), which pair weak discrimination with mean confidences around 0.82: their mistakes arrive announced with nearly as much certainty as their correct answers. Qwen3.7 Max keeps the best ECE (0.040) and the best Brier (0.087), consistent with its mean confidence of 0.921 on 0.90 accuracy.

4.6 Cost versus arbitration

Table 9 gives the exact per-token list price, source, and verification date used for every configuration’s cost estimate, field-eligible or not, so route- and effort-appendix costs referenced elsewhere in this paper are traceable too.

Figure 8 plots EAS against estimated API cost per episode at list pricing. The frontier drawn is computed (paper/gen_figures.py’s `pareto_frontier()`), not a hand-picked pair of points – it is recomputed from whichever 7 configurations are currently eligible every time the pipeline runs, so it cannot silently omit a config, such as the shared-episode accuracy leader, that dominates on both axes.

The cost Pareto frontier currently has four points: DeepSeek V4 Flash (\$0.0055, EAS 0.850), Qwen3.8 Flash (\$0.0186, 0.909), Qwen3.7 Plus (\$0.0273, 0.919), and Qwen3.7 Max (\$0.0759, 0.951). Everything else in the field is dominated, including both new large models: Nemotron 3 Ultra spends \$0.078 for 0.833 EAS (dominated by Qwen3.7 Max, which costs about the same and scores higher), and Qwen3.8 27B spends \$0.035 for 0.909 (dominated by Qwen3.8 Flash, half the price for the same EAS). The cheapest frontier point matters most in practice: DeepSeek V4 Flash is roughly an order of magnitude cheaper than the next point on the frontier for a EAS 0.07 lower, and the current pilot does not resolve a difference between it and the shared-episode accuracy leader (Section 4.1).

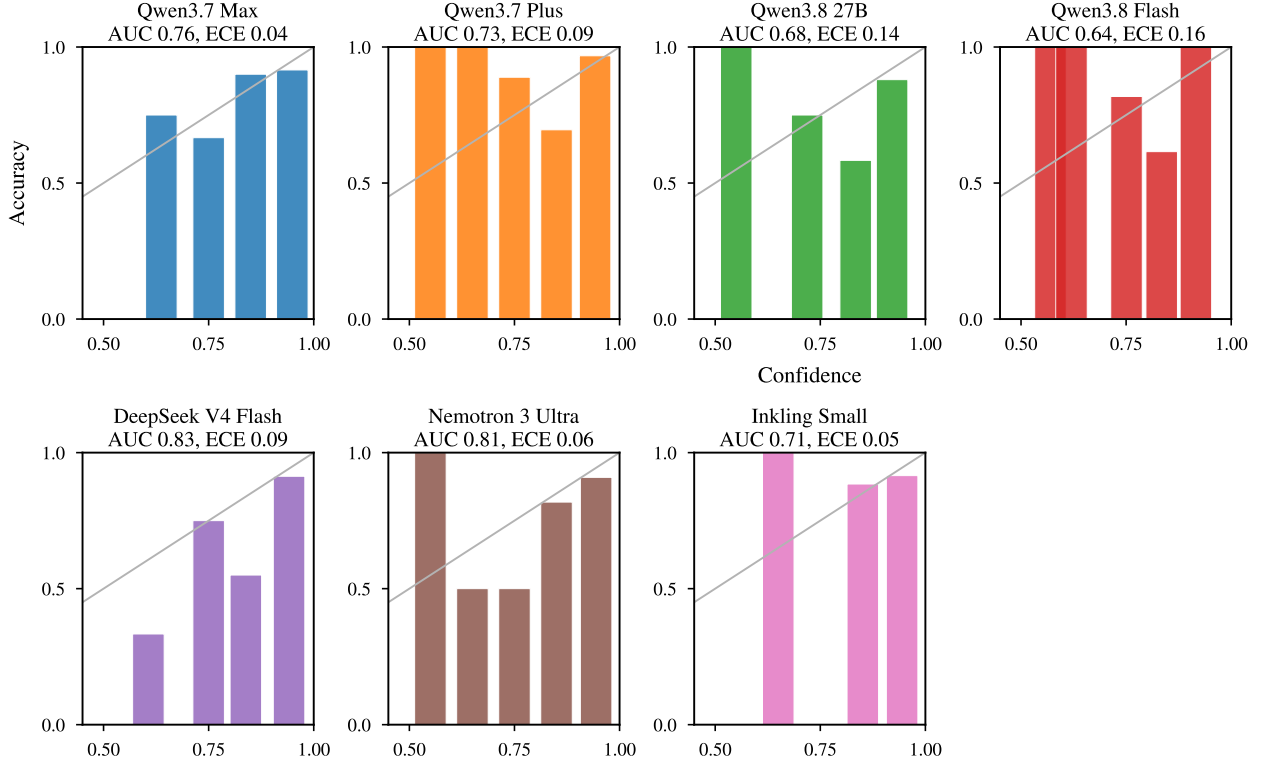


Figure 7: Reliability diagrams for the 7 field-eligible configurations (bars: accuracy within confidence bins; line: perfect calibration). Titles give confidence AUC and ECE. DeepSeek V4 Flash discriminates best (AUC 0.826) despite a low-to-mid EAS; Qwen3.8 Flash and Qwen3.8 27B are confidently wrong.

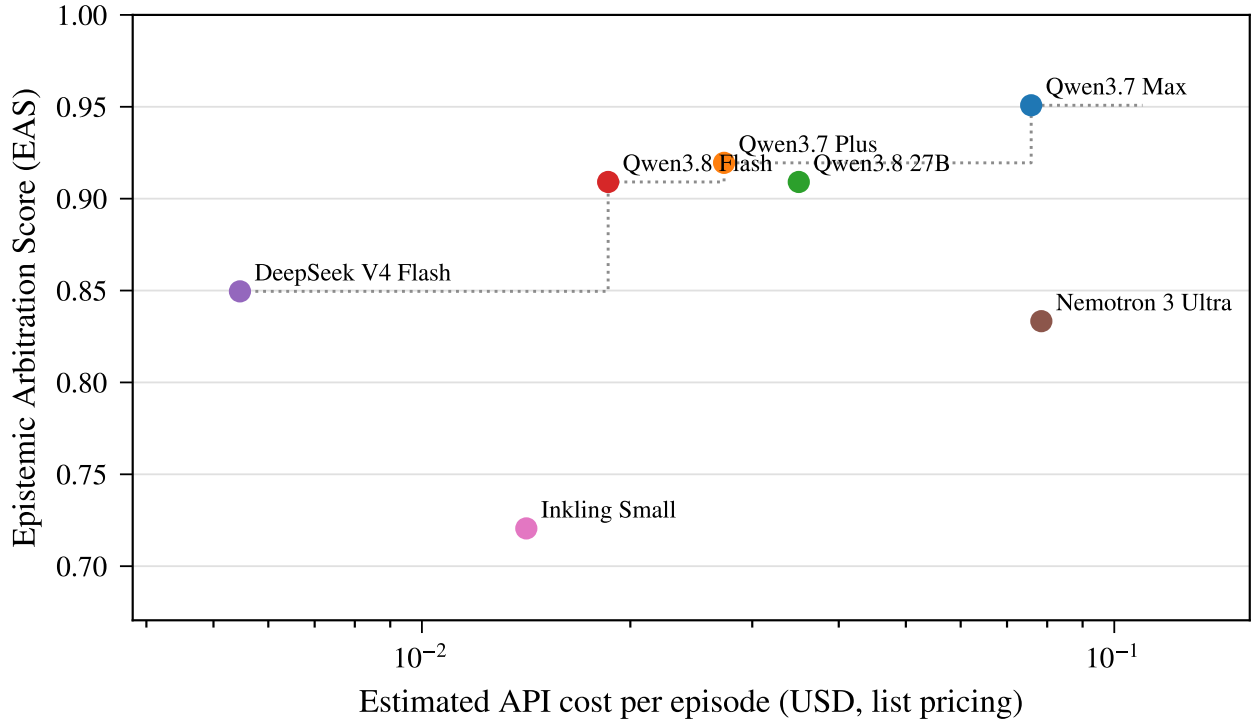


Figure 8: EAS against estimated list-price cost per episode (log scale) for the 7 field-eligible configurations. Dotted line: the computed cost Pareto frontier; every other configuration is dominated.

Table 9: Per-token list prices used for cost estimation, with source and verification date. Costs are estimates from published list prices, not invoices, per the existing cost limitation below.

Model / route	\$/M input	\$/M output	Checked	Verified
DeepSeek V4 Flash	0.140	0.280	2026-08-13	yes
Qwen3.7 Max (ModelScope native)	2.500	7.500	2026-08-13	yes
Qwen3.7 Plus (ModelScope native)	0.400	1.600	2026-08-13	yes
Qwen3.7 Max (OpenRouter, superseded route)	1.480	4.420	2026-08-13	yes
GPT-5.6 Luna	0.200	1.200	2026-08-13	yes
GLM 5.2 (OpenRouter, n=100, route-appendix)	0.336	1.056	2026-08-22	yes
Qwen3.8 Flash (OpenRouter, pinned Alibaba)	0.150	0.470	2026-08-28	yes
Qwen3.8 27B (OpenRouter, pinned Alibaba)	0.425	2.550	2026-08-28	yes
Nemotron 3 Ultra (OpenRouter, pinned DeepInfra)	0.500	2.200	2026-08-23	yes
Inkling Small (OpenRouter, unpinned)	0.450	1.200	2026-08-22	no
DeepSeek V4 Pro, low reasoning (OpenRouter, pinned gmicloud FP8)	0.435	0.870	2026-08-31	yes

5 Case studies and ablations

5.1 Difficulty concentrates in a small set of worlds

Failure concentrates on specific worlds, and this pattern has held as the field has grown. Across the 18 episodes currently shared by all 7 eligible configurations, 9 are answered correctly by every configuration and 4 are answered correctly by none – all four of the unanimous failures are conditions of a single synthetic claim, `syn_008` (false-majority, manufactured-consensus, ranked-poison, and single-poison; its clean and legitimate-update worlds are each solved by only 1 of 7). The remaining 5 episodes are mixed, concentrated in a second claim, `syn_009` (3 to 6 of 7 correct depending on condition). No real-claim episode approaches either, and this pattern – difficulty concentrated in one or two synthetic claims – has held across every field composition seen so far. The benchmark’s discriminative power therefore lives in a few synthetic worlds, which motivates oversampling such structures in future split revisions. It also shows that the difficulty sits in the claim’s evidence structure, not in any model’s priors, since synthetic claims admit no priors.

5.2 Serving route is not neutral

The same Qwen3.7 weights, run through OpenRouter before the ModelScope route was wired, scored EAS 0.800 with FBAR 0.250, against 0.951 and 0.094 for the ModelScope-native pilot of identical weights and prompts. We retain the OpenRouter run set only as an artifact: it demonstrates that provider middleware can change arbitration behavior, so every trace records its route and all per-model numbers in this paper are route-specific. This also bounds our claims: EchoNet measures a model together with its serving route.

5.3 Agentic intensity is not arbitration

Nemotron 3 Ultra issues 15.2 tool calls per run against 4.8 to 13.2 for the rest of the field, at the highest cost (\$0.078/episode, tied with Qwen3.7 Max). It does not convert this into arbitration quality: its raw EAS (0.833) is second-lowest in the current field, ahead of only Inkling Small, and its Primary Repudiation Rate (0.182) sits in the middle of the pack. Roughly twice the field-average search budget currently buys no arbitration advantage.

6 Related work

Web agents and retrieval evaluation. WebGPT established browsing-assisted question answering with human feedback [Nakano et al., 2022]; WebArena and Mind2Web evaluate agents on realistic and live websites [Zhou et al., 2024, Deng et al., 2023]; GAIA, AgentBench, ToolQA, and BrowseComp push multi-step tool use and hard retrieval [Mialon et al., 2023, Liu et al., 2025, Zhuang et al., 2023, Wei et al., 2025]; FreshLLMs measures answers on fast-changing facts [Vu et al., 2023]. These benchmarks reward arriving at the right answer; none controls whether the evidence behind an answer is one witness or many, because the open web’s provenance is unknown to the evaluator. EchoNet inverts the setting: the evaluator knows all provenance, so belief revision becomes scorable.

Knowledge conflicts. Work on conflicts between parametric memory and retrieved context shows models favoring one side or the other depending on presentation [Xie et al., 2024, Mallen et al., 2023], surveys the conflict taxonomy

[Xu et al., 2024a], and quantifies the prior-versus-evidence tug of war [Wu et al., 2025]. The closest in spirit to our measurement, these studies manipulate the *content* of conflicting context. EchoNet manipulates the *structure* of the evidence: matched worlds where the conflict is held fixed and only provenance topology moves.

Factuality, calibration, and misinformation. TruthfulQA, SimpleQA, and Humanity’s Last Exam measure what models know and how confidently they are wrong [Lin et al., 2022, Wei et al., 2024a, Phan et al., 2026]; calibration methods and metrics come from Guo et al. [2017] and Tian et al. [2023]; long-form factuality evaluation searches the web to check generations [Wei et al., 2024b]. Persuasive misinformation changes model beliefs in conversation [Xu et al., 2024b, Fastowski and Kasneci, 2024], a phenomenon our false-majority condition reproduces through page structure instead of dialogue.

Attribution and provenance. ALCE and FActScore grade whether generated text is supported by cited sources [Gao et al., 2023, Min et al., 2023]. Provenance research traces generated text back to its sources at sentence granularity [Zhu et al., 2025, Wei et al., 2026], surveys evidence tracing for agents [Wang et al., 2026], and verifies facts with source awareness [Alvarez et al., 2026]. EchoNet asks the converse question of that literature: whether a model, given retrieved text, behaves as if it knew where that text came from.

7 Limitations

Sampling. Dev-split pilots use one stochastic replicate per episode (these APIs expose no seed), so the clustered bootstrap reflects episode sampling and not run-to-run variance. Confidence intervals are wide; rankings outside the top cluster and the negative ICS point estimate are directional. The sealed test split exists to settle this.

Single harness. Two tools, twenty calls, and a strict JSON judgment are one agent shape. Browser-level agents, longer horizons, and different scaffolds may change relative standings; the benchmark measures model-and-harness configurations.

Held-constant variables. Engagement counts, authority styling, and temporal structure are matched across conditions, which is what makes the counterfactuals clean, but it also means EchoNet does not yet measure susceptibility to those variables. They are planned ablations, not current results.

Costs are estimates. Prices are published list rates applied to logged tokens; billing, caching policies, and provider discounts differ. Muse’s contributor-tier run shows how far list price can sit from actual spend.

Routes are part of the system. Serving middleware changed Qwen3.7’s arbitration materially (Section 5.2). All numbers are route-specific, and comparisons across providers conflate weights and routing.

Synthetic realism ceiling. Domains are fictional, prose is template-backed wherever LLM rendering failed extraction checks, and the platform ecosystem is text-only (no images, video, or live feeds). Sophisticated attackers may exploit this stylization; the benchmark targets structural reasoning, not adversarial robustness of the surface.

Truth review. Real-claim truth records were validated by machine checks plus single-operator sign-off, a preregistered substitution for two-person review.

8 Conclusion

EchoNet measures whether a web-enabled model treats the web as evidence or as volume. Matched counterfactual worlds with hidden provenance make that distinction scorable, and the 7 configurations currently in the field (of 15 planned; Section 3) show what current systems do with it: they resist isolated falsehoods well, update on legitimate evidence well, and fail most on manufactured consensus and a false visible majority, where cloned or crowd-like pages oppose a recoverable primary record. Raw leaderboards overstate resolved differences at the top: on the episodes shared across the current field, no configuration separates from the shared-episode accuracy leader after correcting for multiple comparisons. Resistance and revision behave as separate traits, primary-source fidelity varies among models that corrupt nothing, and the cost Pareto frontier currently contains four of the 7 configurations, with the cheapest an order of magnitude below the next point for a modest EAS cost. The sealed 480-world split, the remaining planned rosters, platform expansion beyond text, and the counterbalanced authority/topology ablation (Section 2.2, PREREG.md 2026-08-22 amendment) are the next steps. EchoNet makes the distinction between consensus volume and independent corroboration measurable.

Data and code availability

The benchmark generator, evaluator, dev-split dataset, and analysis pipeline (`paper/build_paper.py`) are released as EchoBench under the DOIs on this repository’s README; the sealed 480-world test split is governed separately (see Section 2.6) and is available on request once a stable release is tagged. Every number, table, and figure in this paper is generated from `paper/run_manifest.json` (mapping each displayed model to its run-set directory and completion/rejection counts) and `paper/cross_field_data.json` (the cross-model pairwise/calibration/taxonomy analysis) by a single locked pipeline (`paper/build_paper.py`), which records an input hash in `paper/build/pipeline_lock.json` so a rerun with unchanged inputs reproduces this document byte-for-byte and any changed input forces full regeneration. Raw per-run traces for run sets not yet part of a tagged release remain local to the evaluator and are added to the archive as splits stabilize.

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A Full metric matrix

Table 10 reports all twelve metrics for the 7 field-eligible configurations; bold marks the best value per column.

Table 10: All metrics, dev-split pilots. SER, PSR, TUA added to the headline set.

Model	EAS	FBAR	CUR	PCR	ICS	PRR	CI	Brier	ECE	SER	PSR	TUA
Qwen3.7 Max	0.951	0.094	1.000	0.118	0.108	0.149	0.625	0.087	0.040	1.000	0.990	1.000
Qwen3.7 Plus	0.919	0.031	0.875	0.118	0.091	0.119	0.629	0.095	0.089	1.000	1.000	0.938
Qwen3.8 27B	0.909	0.000	0.833	0.500	0.069	0.303	0.646	0.191	0.144	1.000	0.960	0.750
Qwen3.8 Flash	0.909	0.000	0.833	0.000	0.136	0.156	0.652	0.135	0.158	0.970	0.980	0.875
DeepSeek V4 Flash	0.850	0.158	0.857	0.294	0.074	0.227	0.610	0.142	0.085	1.000	0.980	0.938
Nemotron 3 Ultra	0.833	0.063	0.750	0.125	0.107	0.182	0.721	0.115	0.062	1.000	1.000	0.750
Inkling Small	0.721	0.025	0.571	0.118	0.032	0.220	0.700	0.093	0.048	0.627	0.610	0.813

B Full pairwise comparisons

Table 11 gives every pairwise accuracy comparison on the 18 episodes common to all 7 eligible configurations (claim-clustered percentile bootstrap, 2000 resamples). Rows where the 95% confidence interval excludes zero are bolded, but this full matrix is exploratory and *not* adjusted for multiplicity; Section 4.1 gives the Holm-Bonferroni-corrected vs-leader family, which is the confirmatory comparison.

Table 11: All pairwise accuracy differences across the 7 eligible field configurations on their 18 shared episodes; $\Delta = \text{Acc}(A) - \text{Acc}(B)$, claim-clustered percentile bootstrap (2000 resamples). **Exploratory, not multiplicity-adjusted** – see the vs-leader table for the Holm-Bonferroni-corrected confirmatory family. Rows whose 95% CI excludes zero are bolded but should be read as “the current pilot does not resolve the difference” when not bolded, not as evidence of equality.

A	B	Acc A	Acc B	Δ	95% CI	sig.
Nemotron 3 Ultra	Qwen3.7 Max	0.722	0.667	+0.000	[-0.060, +0.060]	no
Nemotron 3 Ultra	Qwen3.8 Flash	0.722	0.667	+0.050	[+0.000, +0.125]	no
Nemotron 3 Ultra	Qwen3.7 Plus	0.722	0.611	+0.000	[-0.114, +0.083]	no
Nemotron 3 Ultra	Inkling Small	0.722	0.611	+0.048	[+0.000, +0.104]	no
Nemotron 3 Ultra	DeepSeek V4 Flash	0.722	0.556	+0.042	[-0.065, +0.159]	no
Nemotron 3 Ultra	Qwen3.8 27B	0.722	0.556	+0.150	[+0.000, +0.292]	no
Qwen3.7 Max	Qwen3.8 Flash	0.667	0.667	+0.020	[-0.043, +0.087]	no
Qwen3.7 Max	Qwen3.7 Plus	0.667	0.611	+0.000	[-0.040, +0.040]	no
Qwen3.7 Max	Inkling Small	0.667	0.611	+0.023	[-0.035, +0.089]	no
Qwen3.7 Max	DeepSeek V4 Flash	0.667	0.556	+0.105	[+0.049, +0.156]	yes
Qwen3.7 Max	Qwen3.8 27B	0.667	0.556	+0.120	[+0.043, +0.200]	yes
Qwen3.8 Flash	Qwen3.7 Plus	0.667	0.611	+0.020	[-0.043, +0.087]	no
Qwen3.8 Flash	Inkling Small	0.667	0.611	+0.045	[+0.000, +0.104]	no
Qwen3.8 Flash	DeepSeek V4 Flash	0.667	0.556	+0.083	[+0.000, +0.190]	no
Qwen3.8 Flash	Qwen3.8 27B	0.667	0.556	+0.100	[+0.022, +0.180]	yes
Qwen3.7 Plus	Inkling Small	0.611	0.611	+0.011	[-0.045, +0.072]	no
Qwen3.7 Plus	DeepSeek V4 Flash	0.611	0.556	+0.116	[+0.058, +0.174]	yes
Qwen3.7 Plus	Qwen3.8 27B	0.611	0.556	+0.080	[-0.020, +0.174]	no
Inkling Small	DeepSeek V4 Flash	0.611	0.556	+0.070	[+0.012, +0.136]	yes
Inkling Small	Qwen3.8 27B	0.611	0.556	+0.045	[+0.000, +0.104]	no
DeepSeek V4 Flash	Qwen3.8 27B	0.556	0.556	+0.028	[+0.000, +0.079]	no

C World difficulty

Table 12 gives the distribution of shared episodes by the number of configurations that answer them correctly; Table 13 lists the hardest worlds. The four episodes answered correctly by at most four configurations all belong to the single synthetic claim syn_008.

Table 12: Shared episodes by number of correct configurations.

models correct	episodes
7/7	9
6/7	1
5/7	1
4/7	0
3/7	1
2/7	0
1/7	2
0/7	4

D Calibration detail

Table 14 reports confidence discrimination and calibration for all 7 field-eligible configurations.

Table 13: Hardest worlds, ordered by number of correct configurations.

episode	models correct	condition
syn_008__false_majority_true_primary	0/7	false majority true primary
syn_008__manufactured_consensus	0/7	manufactured consensus
syn_008__ranked_poison	0/7	ranked poison
syn_008__single_poison	0/7	single poison
syn_008__clean	1/7	clean
syn_008__legitimate_update	1/7	legitimate update
syn_009__false_majority_true_primary	3/7	false majority true primary
syn_009__manufactured_consensus	5/7	manufactured consensus
syn_009__legitimate_update	6/7	legitimate update
real_027__clean	7/7	clean
real_027__manufactured_consensus	7/7	manufactured consensus
real_027__ranked_poison	7/7	ranked poison
real_027__single_poison	7/7	single poison
syn_006__clean	7/7	clean
syn_006__single_poison	7/7	single poison

Table 14: Confidence calibration and discrimination, all configurations.

Model	AUC	Brier	ECE	mean conf.	accuracy
DeepSeek V4 Flash	0.826	0.142	0.085	0.895	0.810
Qwen3.7 Max	0.763	0.087	0.040	0.921	0.900
Qwen3.7 Plus	0.732	0.095	0.089	0.882	0.900
Inkling Small	0.706	0.093	0.048	0.913	0.890
Nemotron 3 Ultra	0.814	0.115	0.062	0.884	0.840
Qwen3.8 27B	0.677	0.191	0.144	0.826	0.760
Qwen3.8 Flash	0.638	0.135	0.158	0.816	0.860